

Han Wang 王晗

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 Peking University, No.5 Yiheyuan Road, Haidian District, Beijing 100871, China

Employment

Kavli Institute for Astronomy and Astrophysics, Peking University (KIAA-PKU)

Beijing, China

Postdoctoral researcher

Jul 2025 - present

- Postdoctoral Supervisor: Lijing Shao (邵立晶)

Education

Sun Yat-sen University (SYSU)

Zhuhai, China

Ph.D. in Astronomy

Sep 2020 - Jun 2025

- Thesis: Space-based Gravitational Wave Data Analysis of Stellar-mass Binary Black Holes: A Multi-band View
- Supervisor: Yi-Ming Hu (胡一鸣)

This thesis focuses on the challenge of detecting gravitational waves from stellar-mass binary black holes in space-based detectors, and uses the idea of archival searches to implement an end-to-end data analysis routine, i.e. from simulated data to detection to parameter estimation. Additionally, the inclusion of eccentricity makes it beneficial to unveil the formation mechanisms of such systems.

University of Portsmouth

Portsmouth, UK

Visiting Ph.D. Student

Oct 2023 - Oct 2024

- Supervisor: Ian Harry

Huazhong University of Science and Technology (HUST)

Wuhan, China

B.S. in Physics

Sep 2016 - Jun 2020

- Thesis: Analyses of Laser Propagation Noises for TianQin Gravitational Wave Observatory
- Supervisors: Yan Wang (王炎) and Wei Su (苏威)

This thesis focuses on how global magnetosphere will introduces additional noise to the TianQin Gravitational Wave Observatory and estimates its impact using MHD simulation data.

University of California, Berkeley

Berkeley, CA, USA

Berkeley International Study Program

Jan 2019 - May 2019

Selected Awards and Scholarships

Honors Graduate of SYSU

2025

Top distinction awarded to the top 5% of graduates at SYSU

Outstanding Dissertation Award, School of Physics and Astronomy, SYSU

2025

School-level recognition for an excellent doctoral dissertation highly commended by blind reviewers

National Scholarship

2024

Top honor awarded to outstanding students for exceptional academic achievements, leadership, and contributions to society

State Scholarship Fund, China Scholarship Council

Oct 2023 - Oct 2024

Awarded to support overseas study, in this case a visit to the University of Portsmouth, following a rigorous academic evaluation process

TianQin Jianxing Outstanding Postgraduate Scholarship

2022 & 2023

Endowed scholarship for outstanding postgraduates contributing to the TianQin Gravitational Wave Detection Mission

The Postgraduate Skills Competition of TianQin Research Center

1st in 2021

Recognition awarded to postgraduates for producing excellent scientific outreach videos on gravitational-wave topics

& 2nd in 2022

Outstanding Graduate of HUST <i>Top honor awarded to graduating undergraduates at HUST</i>	2020
National Astronomical Observatories Scholarship, Chinese Academy of Sciences <i>Awarded to outstanding undergraduates nationwide who had chosen astronomy as their research direction</i>	2017 & 2019
Merit Student of HUST <i>Recognition awarded to undergraduates for excellence in both academic performance and public service</i>	2019
Outstanding Student Cadre of HUST <i>Recognition awarded to outstanding undergraduate leaders actively involved in student associations or organizations</i>	2017 & 2018

Conferences and Presentations

Conferences:

2 nd Graduate Academic Forum of the School of Physics and Astronomy (Gewu Forum), <i>Sun Yat-sen University</i> , Zhuhai, China - Talk: 1st Prize & Audience Choice Award	May 2025
2025 Annual Meeting of the Gravity and Relativistic Astrophysics Division of the Chinese Physical Society, <i>Kunming University of Science and Technology</i> , Kunming, China - Talk	Apr 2025
245 th Meeting of the American Astronomical Society, <i>National Harbor</i> , Maryland, USA - Talk	Jan 2025
2024 Annual Meeting of the TianQin Research Center, <i>TianQin Research Center</i> , Zhuhai, China - Talk: Best Student Talk Award	Jan 2025
2024 Annual Meeting of the Chinese Astronomical Society, <i>Zhejiang University</i> , Hangzhou, China - Poster	Oct 2024
15 th LISA Symposium, <i>University College Dublin</i> , Dublin, Ireland - Poster 🔗	Jul 2024
Faculty of Technology Research and Innovation Conference, <i>University of Portsmouth</i> , Portsmouth, UK - Poster	Jun 2024
24 th BritGrav meeting, <i>Queen Mary University of London</i> , London, UK - Talk	Apr 2024
1 st Symposium on Gravitational Wave Astronomy in the Audio Band, <i>Beijing Normal University</i> , Zhuhai, China - Remote Talk	Mar 2024
2023 Annual Meeting of the Gravity and Relativistic Astrophysics Division of the Chinese Physical Society, <i>Chongqing University</i> , Chongqing, China - Talk	Apr 2023

Selected Seminars:

Kavli Institute for Astronomy and Astrophysics, <i>Peking University</i> , Beijing, China - Invited Talk	Apr 2025
LISA Ground Segment Meet Up, <i>University of Birmingham</i> , Birmingham, UK	Jan 2024
SYSU-PKU Bilateral Seminar on Gravitational Wave Astronomy, <i>TianQin Research Center</i> , Zhuhai, China	Mar 2023
Research Tools and Techniques Seminar, <i>TianQin Research Center</i> , Zhuhai, China	Dec 2021

Outreach and Services

Lead developer of [TianQinSYSU/GWSpace](#) [↗](#)

Illustration development of TianQin Gravitational Wave Detection Mission

Outreach videos on gravitational waves [In Chinese] [link1](#) [↗](#) [link2](#) [↗](#)

Volunteer of [Stargazing at Portsmouth Historic Dockyard](#) [↗](#)

Jan 2024

Leader of Astronomy Enthusiasts Group of TianQin Research Center

2020 - 2022

Organizing sidewalk astronomy observation and other outreach events.

Lab manager of the Innovative Base for Physics Experiment (IBPE) of HUST

2017 - 2018

Responsible for lab planning, procurement, daily maintenance, public outreach, etc.

IBPE is a free platform that encourages undergraduates to engage in physics-related self-study and self-research.

Vice President of the Astronomy Enthusiasts Association of HUST

2017 - 2018

Core management of the Astronomy Enthusiasts Association of HUST

2016 - 2019

Organizing sidewalk astronomy observation and camping & stargazing trips, inviting talks by astronomy experts, writing scientific outreach content for members, live-streaming astronomical events (e.g., total lunar eclipses), giving astronomy outreach lectures at local high schools, etc.

Teaching

Teaching Assistant of

Methods in Mathematical Physics

Fall 2022

Fundamentals of Physics II

Fall 2021

Thermodynamics and Statistical Physics

Spring 2020

Publications

- Han Wang**, Michael J. Williams, Ian Harry, Yi-Ming Hu. “Archival Inference for Eccentric Stellar-Mass Binary Black Holes in Space-Based Gravitational Wave Observations”. In: *arXiv e-prints* (2025). arXiv: 2510.07174 [astro-ph.HE] [↗](#). URL: <https://arxiv.org/abs/2510.07174>.
- Han Wang**, Ian Harry, Alexander Nitz, Yi-Ming Hu. “Space-based gravitational wave observatories will be able to use eccentricity to unveil stellar-mass binary black hole formation”. In: *Phys. Rev. D* 109.6, 063029 (Mar. 2024), p. 063029. doi: 10.1103/PhysRevD.109.063029 [↗](#). arXiv: 2304.10340 [astro-ph.HE] [↗](#).
- En-Kun Li*, **Han Wang***, Hong-Yu Chen, Huimin Fan, Ya-Nan Li, et al. “GWSpace: a multi-mission science data simulator for space-based gravitational wave detection”. In: *Class. Quant. Grav.* 42.16 (2025), p. 165005. doi: 10.1088/1361-6382/adf409 [↗](#). arXiv: 2309.15020 [gr-qc] [↗](#).
- Hong-Yu Chen, **Han Wang**, En-Kun Li, Yi-Ming Hu. “Signal-to-noise ratio analytic formulae of the inspiral binary black holes in TianQin”. In: *Class. Quant. Grav.* 42.15 (2025), p. 155010. doi: 10.1088/1361-6382/adeed9 [↗](#). arXiv: 2410.19401 [astro-ph.GA] [↗](#).
- En-Kun Li, Hong-Yu Chen, Ya-Nan Li, Zhi-Yuan Li, **Han Wang**, et al. “Data Analysis of Space-borne Gravitational Wave Missions (in Chinese)”. In: *Sci. Sin.-Phys. Mech. Astron.* 55 (2025), p. 230406. doi: 10.1360/SSPMA-2024-0357 [↗](#). URL: <https://www.sciengine.com/SSPMA/doi/10.1360/SSPMA-2024-0357>.
- En-Kun Li, Shuai Liu, Alejandro Torres-Orjuela, Xian Chen, Kohei Inayoshi, et al. (incl. **Han Wang**). “Gravitational wave astronomy with TianQin”. In: *Rept. Prog. Phys.* 88.5 (Sept. 2025), p. 056901. doi: 10.1088/1361-6633/adc9be [↗](#). arXiv: 2409.19665 [astro-ph.GA] [↗](#).
- Hanzhang Wang, Shuai Liu, **Han Wang**, Chen Hong-Yu, Long Wang, Yi-Ming Hu. “Detection for Intermediate-mass Binary Black Holes in Population III Star Clusters with TianQin”. In: *Astron. Astrophys.* 697 (Mar. 2025), A49. doi: 10.1051/0004-6361/202453615 [↗](#). arXiv: 2503.15907 [astro-ph.GA] [↗](#).
- Wei Su, Yan Wang, Chen Zhou, Lingfeng Lu, Ze-Bing Zhou, et al. (incl. **Han Wang**). “Analyses of Laser Propagation Noises for TianQin Gravitational Wave Observatory Based on the Global Magnetosphere MHD Simulations”. In: *Astrophys. J.* 914.2, 139 (June 2021), p. 139. doi: 10.3847/1538-4357/abfc49 [↗](#). arXiv: 2102.10574 [astro-ph.SR] [↗](#).